

# Erdős Problem #679

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| EP-679 | L1 - Tractable | Partiellement résolu | Number Theory |
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## Énoncé original (anglais, reproduction mot pour mot)

Let  $\epsilon > 0$  and  $\omega(n)$  count the number of distinct prime factors of  $n$ . Are there infinitely many values of  $n$  such that  $\omega(n-k) < (1+\epsilon) \frac{\log k}{\log \log k}$  for all  $k < n$  which are sufficiently large depending on  $\epsilon$  only?

Can one show the stronger version with  $\omega(n-k) < \frac{\log k}{\log \log k} + O(1)$  is false?

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